

Scala For DS

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Class : SYIT	Assignment Date : 29/09/2026

Practical no.30

INPUT:

```
import breeze.linalg._

object PRACTICAL extends App {

    val data = DenseVector(1.0, 2.0, 3.0, 10.0, 11.0, 12.0)

    var center1 = 2.0
    var center2 = 10.0

    // Repeat clustering
    for (i <- 1 to 5) {

        val cluster1 = data.toArray.filter(x =>
            | math.abs(x - center1) < math.abs(x - center2)
            | )

        val cluster2 = data.toArray.filter(x =>
            | math.abs(x - center2) <= math.abs(x - center1)
            | )

        center1 = cluster1.sum / cluster1.length
        center2 = cluster2.sum / cluster2.length
    }
}
```

```

println("Cluster 1:")
println(data.toArray.filter(x =>
|   math.abs(x - center1) < math.abs(x - center2)
|).mkString(", "))

println("Cluster 2:")
println(data.toArray.filter(x =>
|   math.abs(x - center2) <= math.abs(x - center1)
|).mkString(", "))

// Display centers
println("Center 1 = " + center1)
println("Center 2 = " + center2)
println("S007 DHRUV")
}

```

OUTPUT:

```

C:\Users\jiawa\.jdk\ms-17.0.20.1\bin
Cluster 1:
1.0, 2.0, 3.0
Cluster 2:
10.0, 11.0, 12.0
Center 1 = 2.0
Center 2 = 11.0
S007 DHRUV

Process finished with exit code 0

```